

Chapter 1: Quadratic Functions

NOTES

1.1 Maximum and Minimum values

① Completed square Form

$$f(x) = a(x-h)^2 + k$$

When $a > 0$, minimum value = k

When $a < 0$, maximum value = k

How to complete the square
from $ax^2 + bx + c$:

$$ax^2 + bx + c$$

↓ factorise a from x^2 and x

$$a(x^2 + \frac{b}{a}x) + c$$

↓ add constant term squared

$$a[x^2 + \frac{b}{a}x + (\frac{b}{2a})^2 - (\frac{b}{2a})^2] + c$$

$$a(x + \frac{b}{2a})^2 - \frac{b^2}{4a} + c \quad (\text{completed square form})$$

1.2 Graphical Representation of Quadratic Functions

② Turning point

$$y = a(x-h)^2 + k$$

$$\text{Turning point} = (h, k)$$

③ Discriminant

$$\text{discriminant} = b^2 - 4ac$$

$$b^2 - 4ac < 0: \text{no real roots}$$

$$b^2 - 4ac = 0: \text{two real and repeated roots}$$

$$b^2 - 4ac > 0: \text{two real and distinct roots}$$

④ Always positive / negative

$$1. \text{discriminant} < 0 \text{ and } a > 0:$$

→ always positive

$$2. \text{discriminant} < 0 \text{ and } a < 0:$$

→ always negative

QUESTION TYPES

1.1 Maximum and Minimum values

1. Find all possible real values of a quadratic function. ①

Step 1: Complete the square

Step 2: If $a > 0$:

Since $(x-h)^2 \geq 0$ for all real values of x ,

$$a(x-h)^2 \geq 0$$

$$a(x-h)^2 + k \geq k$$

$$f(x) \geq k$$

Step 3: If $a < 0$

Since $(x-h)^2 \geq 0$ for all real values of x ,

$$a(x-h)^2 \leq 0$$

$$a(x-h)^2 + k \leq k$$

$$f(x) \leq k$$

2. Show that function is negative / positive for all real x . ①

Step 1: complete the square

Step 2: copy [1]:

$$f(x) \geq k > 0 \rightarrow \text{true}$$

$$f(x) \leq k < 0 \rightarrow \text{true}$$

1.2 Graphical Representation of Quadratic Functions

3. Find values of unknowns / range of values of unknowns such that graph meets x -axis at zero / one / two points.

Step 1: Set discriminant:

$$\text{Zero} \rightarrow \text{discriminant} < 0$$

$$\text{One} \rightarrow \text{discriminant} = 0$$

$$\text{Two} \rightarrow \text{discriminant} > 0$$

4. Show that graph lies completely above / below the x -axis.

Step 1: Find discriminant.

(if discriminant is a quadratic function show that $d < 0$) [1]

Step 2: check coeff of x^2 .

Step 3: Explain.

Since discriminant < 0 and $a > 0$
OR $a < 0$, $f(x)$ is always negative / positive.

5. Find range of unknown values if graph is always positive / negative.

Step 1: For graph to be always positive / negative, discriminant < 0 and $a > 0$ OR $a < 0$.

Step 2: Simplify to find range of unknown value.

1.3 Applications of quadratic functions

6. Determine if graph passes through a point.

Step 1: Substitute x -coordinate into eqn of graph.

7. Given turning point and intercepts, find quadratic function in the form: $y = a(x-p)(x-q)$.

Step 1: Identify x -coordinates of x -intercepts (substitute as p and q).

Step 2: Substitute coordinates of turning point to find value of a .

8. Given turning point and intercepts, find quadratic function in the form: $y = a(x-h)^2 + k$.

Step 1: Substitute coordinates of turning point to find h and k . (h, k)

Step 2: Substitute coordinates of x -intercept to find value of a .

Chapter 2: Quadratic Equations and Inequalities

NOTES

2.1 Simultaneous Equations

① Solutions of simultaneous equations
 solutions $\rightarrow x_1, y_1$ and x_2, y_2

Points of intersection of the line and the curve are (x_1, y_1) and (x_2, y_2) .

2.2 Nature of Roots of a Quadratic Equation

② Discriminant and points of intersection

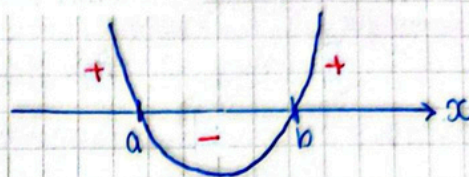
discriminant $< 0 \rightarrow 0$ pts of intersection

discriminant $= 0 \rightarrow 1$ pt of intersection

discriminant $> 0 \rightarrow 2$ pts of intersection

2.3 Quadratic Inequalities

③ Solving quadratic inequalities - graph



QUESTION TYPES

2.1 Simultaneous Equations

1. Solve simultaneous equations/
 points of intersection.

Step 1: Make y the subject in the linear equation.

Step 2: Substitute value of y and solve quadratic in x .

Step 3: Substitute values of x into linear equation to find values of y . ①

2.2 Nature of Roots of a Quadratic Equation

2. Determine if line intersects curve/number of points of intersection

Step 1: Substitute equation of line into equation of curve to obtain quadratic equation in x .

Step 2: Equate discriminant. ②
 discriminant $= \dots = n$ (< 0 / $= 0$ / > 0)

3. Find unknowns in equations given points of intersection.

Step 1: Substitute equation of line into equation of curve to obtain quadratic equation in x .

Step 2: Explain + discriminant ②
 Since line intersects curve at $(0, 1/2)$ points,
 discriminant ...
 (solve for unknown)

2.3 Quadratic Inequalities

4. Solve quadratic inequalities

Step 1: Ensure that coefficient of x^2 is positive.

Step 2: Factorise if possible, to get points a and b . ③

Step 3: Otherwise, use quadratic formula to get points a and b . ③

Step 4: Draw curve and write the inequality solution. ③

5. Quadratic inequalities - given solution and coefficient of x^2 , form the inequality.

Step 1: Draw graph to visualise. ③

Step 2: Determine factorised form and whether it is < 0 or > 0 (work backwards)

Step 3: Expand and compare coefficients.

Additional Mathematics

Chapter 3: Surds

NOTES

3.1 Manipulating Surds

① Properties of surds

1. $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$
2. $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
3. $\sqrt{a} \times \sqrt{a} = a$

② Rationalising the denominator of a fraction

denominator: conjugate:

1. $\sqrt{k}$ $\sqrt{k}$
2. $a + b\sqrt{k}$ $a - b\sqrt{k}$
3. $a\sqrt{h} + b\sqrt{k}$ $a\sqrt{h} - b\sqrt{k}$

(Multiply the denominator and the conjugate to rationalise)

3.2 Solving Equations Involving Surds

③ Solving Equations involving surds - Check
When there is more than one solution,
substitute solution into equation to
check if it is valid.

④ Equations in rational and irrational
terms with unknowns

If $a + b\sqrt{k} = c + d\sqrt{k}$, where a, b, c, d
are rational and $\sqrt{k}$ is irrational,
then $a = c$
 $b = d$

QUESTION TYPES

3.1 Manipulating Surds

1. Simplifying surds to combine like terms

Step 1: Find a factor of the number that is a square number.

Step 2: Use $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ to factor out square number. ①

2. Rationalise the denominator of a fraction

Step 1: Identify conjugate surd. ②

Step 2: Multiply fraction with $\frac{\text{conjugate}}{\text{conjugate}}$ then simplify.

3. Given area of rectangle and width, find the length.

Step 1: Define variable for length.

Step 2: Length $\times$ Width = Area

Step 3: Rearrange terms and rationalise denominator. ②

3.2 Solving Equations Involving Surds

4. Solving equations where unknown variable is in the radical.

Step 1: Ensure that there is ONLY one radical on LHS and/or RHS.

Step 2: Square both sides. If there is a constant term, expand $(a+b)^2$ and isolate radical. Then square both sides again.

Step 3: Expand and simplify. ③

5. Solving equations where unknown is outside radical.

Step 1: Expand and simplify.

Rearrange terms to form $a + b\sqrt{k} = c + d\sqrt{k}$

Step 2: Compare terms using property. ④

Comparing rational terms,
Comparing irrational terms,

Additional Mathematics

Chapter 4: Polynomials and Partial Fractions

NOTES

4.1 Polynomials and Identities

① Identities - finding unknowns in identities

Method 1 - Equate coefficients

Method 2 - substitute values of x into the identity

② Division of polynomials

 $\text{dividend} = \text{divisor} \times \text{quotient} + \text{remainder}$

degree of remainder MUST BE less than degree of divisor

$$P(x) = D(x)Q(x) + R(x)$$

4.2 The Remainder and Factor Theorems

③ The Remainder Theorem

$$f(x) \div (ax+b) \rightarrow \text{Remainder: } f\left(-\frac{b}{a}\right)$$

$$f(x) \div (x-a) \rightarrow \text{Remainder: } f(a)$$

(Set divisor to zero and move everything to RHS, that is the value to substitute in)

④ The Factor Theorem

a. If $x-a$ is a factor of $f(x)$, $f(a) = 0$ b. If $f(a) = 0$, $x-a$ is a factor of $f(x)$
 $\rightarrow (f(x) \text{ is divisible by } x-a)$

4.3 Cubic Polynomials and Equations

⑤ Sum and difference of cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

4.4 Partial Fractions

⑥ Proper vs Improper Fractions

$$\frac{f(x)}{g(x)}$$

Proper if degree of $f(x) < \text{degree of } g(x)$ Improper if degree of $f(x) \geq \text{degree of } g(x)$

⑦ Partial Fraction Decomposition

Denominator

Partial Fraction

1. $(ax+b)(cx+d) \quad \frac{A}{ax+b} + \frac{B}{cx+d}$

2. $(ax+b)^2 \quad \frac{A}{ax+b} + \frac{B}{(ax+b)^2}$

3. $x^2 + C \quad \frac{Ax+B}{x^2+C}$

QUESTION TYPES

4.1 Polynomials and Identities

1. Find unknowns in identities.

Method 1 - Equate coefficients

 $\rightarrow$ use if in expanded form**Equating the coefficients of (term).**Method 2 - substitute values of x $\rightarrow$ use if in factorised form**Substituting $x = n$.**

2. Division of Polynomials.

(quotient)

$$(\text{divisor}) \overline{) ax^n + bx^{n-1} \dots}$$

(remainder)

★ If the polynomial skips powers

e.g. $x^3 + x^{n-2}$ add a $0x^{n-1}$ term when doing long division.

4.2 The Remainder and Factor Theorems

3. Find unknowns in $f(x)$ given

remainder when divided by (divisor)

Step 1: **By remainder theorem, ③**

$$f(\text{value}) = \text{remainder}$$

Step 2: $\therefore$ solve for unknown4. Find unknowns in $f(x)$ given that

two divisors give the same remainder

Step 1: Equate and expand.

$$f(m) = f(n)$$

Step 2: Simplify to solve for unknown

5. Given the remainders of $f(x)$ when divided by two linear divisors, find remainder when $f(x)$ is divided by quadratic divisor.

Step 1: Let the remainder when $f(x)$ is divided by (quadratic divisor) be $ax+b$.

Step 2: Equate $f(x)$ as
dividend = divisor $\times$ quotient + remainder.

Let $f(x) = (x-m)(x-n)Q(x) + ax+b$

Step 3: $f(m)$ - equation 1

$f(n)$ - equation 2

$$a(m)+b = (1)$$

$$a(n)+b = (2)$$

Step 4: Solve simultaneous equations for a and b .

6. Determine if a linear divisor is a factor of $f(x)$ / exactly divisible by $f(x)$.

Step 1: $f(n) = \dots$

Step 2: By factor theorem, $x-n$ is a factor of $f(x)$.

7. Finding two unknowns in $f(x)$ given quadratic factor.

Step 1: Factorise quadratic factor.

Step 2: $f(x)$ is exactly divisible by $(x-m)$ and $(x-n)$.

Step 3: Use factor theorem to form two equations.

$$f(m) = 0 \dots (1)$$

$$f(n) = 0 \dots (2)$$

Step 4: Solve simultaneous equations for unknowns.

4.3 Cubic Polynomials and Equations

8. Factorising/solving cubic equations

Step 1: Substitute random values into $f(x)$ until $f(x) = 0$.

$\rightarrow$ Try $x = 0, 1, -1, 2, -2, 0.5, -0.5$ etc.

Step 2: By factor theorem, $x-n$ is a factor of $f(x)$.

Step 3: Let $P(x) = (x-n)(ax^2+bx+c)$
 $\rightarrow$ Identify a and c by observing first & last term.

Step 4: Find value of b by equating coefficients of x .

Step 5: Factorise (ax^2+bx+c) if possible.

4.4 Partial Fractions

9. Partial Fractions - Improper Fraction

Step 1: Long division

Express the fraction as such.

$$\text{Fraction} = \text{Quotient} + \frac{\text{Remainder}}{\text{Divisor}}$$

Step 2: Partial Fraction on
Remainder
Divisor

Step 3: Final answer:

$$\text{Fraction} = \text{Quotient} + (\text{partial fractions})$$

Additional Mathematics

Chapter 5: Exponential and Logarithmic Functions

NOTES

5.1 Exponential Equations

① Solving exponential equations:

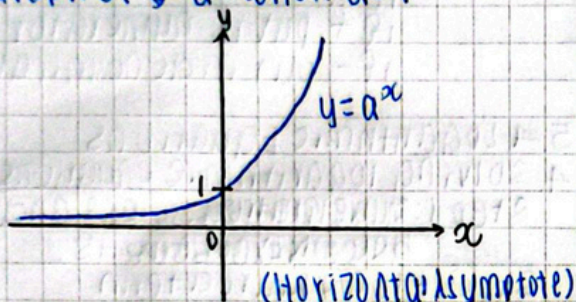
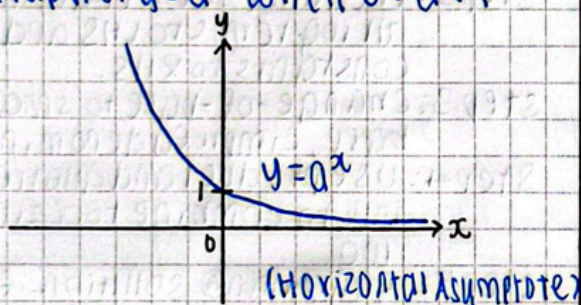
property

$$a^m = a^n \rightarrow m = n$$

② Exponential functions range

$$a^x > 0 \text{ for all real } x.$$

5.2 Graphs and Applications of Exponential Functions

③ Graph of $y = a^x$ when $a > 1$ ④ Graph of $y = a^x$ when $0 < a < 1$ ⑤ Exponential Functions in the form $y = ka^x + c$

$$y\text{-intercept} \rightarrow k + c$$

$$\text{Asymptote} \rightarrow y = c$$

When $k < 0$, graph flips (below $y = c$)

5.3 Logarithms

⑥ Logarithmic Form

$$y = a^x \leftrightarrow \log_a y = x$$

⑦ Logarithmic Functions range

For $\log_a b$ to be defined,

$$a > 0, a \neq 1$$

$$b > 0$$

⑧ Common and Natural Log

$$\text{Common log: } \log_{10} x \rightarrow \lg x$$

$$\text{Natural log: } \log_e x \rightarrow \ln x$$

⑨ Special Log values

$$\log_a a = 1$$

$$\log_a 1 = 0$$

⑩ Logarithm Laws

$$1. \text{Power Law} \rightarrow \log_a x^r = r \log_a x$$

$$2. \text{Product Law} \rightarrow \log_a (xy) = \log_a x + \log_a y$$

$$3. \text{Quotient Law} \rightarrow \log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$4. \text{Change-of-Base Law:}$$

$$\log_a b = \frac{\log_c b}{\log_c a} \text{ AND } \log_c a \times \log_a b = \log_c b$$

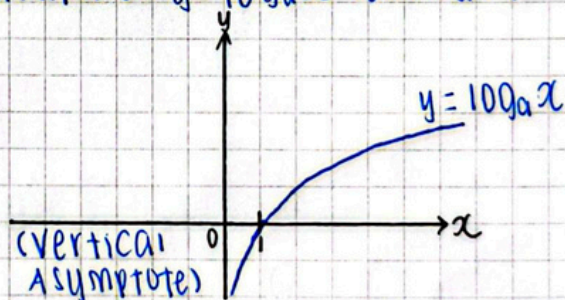
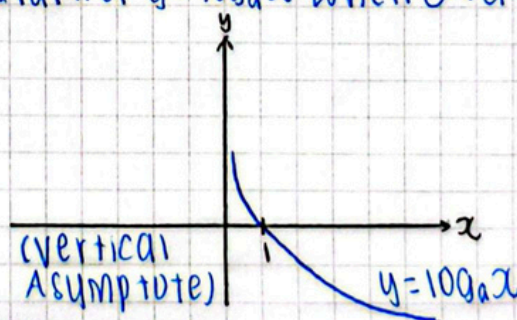
5.4 Logarithmic Equations

⑪ Solving logarithmic equations:

property

$$\log_a M = \log_a N \rightarrow M = N$$

5.5 Graphs and Applications of Logarithmic Functions

⑫ Graph of $y = \log_a x$ when $a > 1$ ⑬ Graph of $y = \log_a x$ when $0 < a < 1$ 

QUESTION TYPES

5.1 Exponential Equations

1. Solving Exponential Equations

Step 1: Rewrite everything using the same base.

Step 2: Substitute $u = a^x$ and simplify into quadratic/linear.

Step 3: Replace the substitution and reject if $a^x \leq 0$. ②

Step 4: Solve for x using ① or using $\lg/\ln$ [5]

5.2 Graphs and Applications of Exponential Functions

2. Explain why $y = ka^x + c > c$ for all real x .

Since $a^x > 0$ for all real x ,

$ka^x > 0$ OR $ka^x < 0$

$ka^x + c > c$ OR $ka^x + c < c$

5.3 Logarithms

3. Simplify complicated logarithmic expressions into a single $\lg$.

Step 1: Convert constants.

$n \rightarrow n \lg_a a$, where a is the smallest base.

Step 2: Power law: remove constants in front of terms:

$n \lg_a b \rightarrow \lg_a b^n$

Step 3: Change-of-Base to smallest base. Simplify denominators.

Step 4: Use product and quotient law to combine to a single $\lg$:

if + put in numerator.

if - put in denominator

5.4 Logarithmic Equations

4. Solving logarithmic equations

Step 1: Substitution:

Use ONLY when:

- $\lg$ squared term

- $\lg$ in denominator (Move to step 5)

Step 2: If cannot substitute, move all $\lg$ terms to LHS and constants to RHS.

Step 3: Change-of-base to smallest base, simplify denominators.

Step 4: Use product and quotient law to combine to a single $\lg$.

Step 5: Convert: ⑥

$\lg_a b = c \rightarrow b = a^c$

Step 6: Reject solutions. ⑦

5. Solutions of Exponential Equations using Logarithms ($a^x = c$)

Step 1: Take $\ln$ on both sides

$\ln a^x = \ln c$

Step 2: Power law: Bring down x :

$x \ln a = \ln c$

Step 3: Solve for x in calculator:

$x = \frac{\ln c}{\ln a}$